Customer Churn Prediction

This project focuses on predicting whether a telecom customer will leave the service or continue using it. Machine Learning classification techniques are applied to analyze customer behavior, train predictive models, and identify factors that influence customer churn.

Environment Setup

1. **Clone the repository or download the project files.**

2. **Create a virtual environment (recommended):**

```
python -m venv venv
```

1. **Activate the virtual environment:**

2. On Windows:

   ```
   .\venv\Scripts\activate
   ```

3. On macOS/Linux:

   ```
   source venv/bin/activate
   ```

4. **Install the required dependencies:**

```
pip install -r requirements.txt
```

Dataset

The project uses the **Telco Customer Churn Dataset** to analyze customer information and predict churn behavior.

The dataset contains customer details such as:

- Customer demographics
- Account information
- Service usage details
- Billing information
- Churn status

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn

Machine Learning Models

The following classification models were implemented:

1. **Logistic Regression**
2. **Random Forest Classifier**

Project Workflow

The project follows these steps:

1. Data Loading
2. Data Cleaning and Preprocessing
3. Exploratory Data Analysis (EDA)
4. Feature Encoding
5. Train-Test Split
6. Model Training
7. Model Evaluation
8. Feature Importance Analysis
9. Model Saving

Evaluation Metrics

The models were evaluated using the following performance metrics:

- Accuracy Score
- Precision Score
- Recall Score
- F1 Score
- ROC-AUC Score

How to Run

1. **Start Jupyter Notebook or JupyterLab:**

```
jupyter notebook
```

or

```
jupyter lab
```

1. **Open and run the Customer Churn Prediction notebook.**

2. The trained model will be saved after evaluation for future predictions.

Model Performance

Both Logistic Regression and Random Forest models were trained and evaluated.

The Random Forest Classifier achieved better performance and was selected as the final model for customer churn prediction.

Output

The project provides:

- Trained machine learning model file (`.pkl`)
- Performance evaluation results
- Customer churn predictions
- Feature importance analysis to understand key churn factors